

Circuit Simulation Lab:28

Design of FIFO

The purpose of this experiment is to design a 8 bit FIFO. FIFO (first-in-first-out) are used to for serial transfer of information whenever there is a difference of Transfer rate. The Transfer rate may differ due to difference in number of ports, frequency or data-width between source and destination.

CIDE:

```
module fifo_8bit(
input clk,
input rst,

input wr_en,
input rd_en,

input [7:0] data_in,

output reg [7:0] data_out,
output fifo_full,
output fifo_empty
);

reg [7:0] mem [0:7];

reg [2:0] wr_ptr;
reg [2:0] rd_ptr;

reg [3:0] count;

assign fifo_empty = (count == 0);
assign fifo_full  = (count == 8);

always @(posedge clk)
begin

if(rst)
begin
    wr_ptr  <= 0;
    rd_ptr  <= 0;
    count   <= 0;

```

```

        data_out <= 0;
    end

    else
    begin

        // WRITE
        if(wr_en && !fifo_full)
        begin
            mem[wr_ptr] <= data_in;
            wr_ptr <= wr_ptr + 1;
            count <= count + 1;
        end

        // READ
        if(rd_en && !fifo_empty)
        begin
            data_out <= mem[rd_ptr];
            rd_ptr <= rd_ptr + 1;
            count <= count - 1;
        end

    end

end

end

Endmodule

TESTBENCH
module tb_fifo_8bit;

    reg clk;
    reg rst;

    reg wr_en;
    reg rd_en;

    reg [7:0] data_in;

    wire [7:0] data_out;

    wire fifo_full;
    wire fifo_empty;

```

```

fifo_8bit dut(
.clk(clk),
.rst(rst),
.wr_en(wr_en),
.rd_en(rd_en),
.data_in(data_in),
.data_out(data_out),
.fifo_full(fifo_full),
.fifo_empty(fifo_empty)
);

always #10 clk = ~clk;

initial begin

$dumpfile("fifo.vcd");
$dumpvars(0,tb_fifo_8bit);

clk = 0;
rst = 1;

wr_en = 0;
rd_en = 0;
data_in = 0;

#20;
rst = 0;


// WRITE DATA

wr_en = 1;

data_in = 8'h11; #20;
data_in = 8'h22; #20;
data_in = 8'h33; #20;
data_in = 8'h44; #20;

wr_en = 0;


// READ DATA

rd_en = 1;

```

```

#20;
#20;
#20;
#20;

rd_en = 0;

#40;

$finish;

end

initial
begin

$monitor(
"time=%0t din=%h dout=%h full=%b empty=%b",
$time,
data_in,
data_out,
fifo_full,
fifo_empty
);

end

Endmodule

```

```

vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_28$ iverilog -o out
fifo_8bit.v tb_fifo_8bit.v
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_28$ vvp out
VCD info: dumpfile fifo.vcd opened for output.
time=0 din=00 dout=xx full=x empty=x
time=10 din=00 dout=00 full=0 empty=1
time=20 din=11 dout=00 full=0 empty=1
time=30 din=11 dout=00 full=0 empty=0
time=40 din=22 dout=00 full=0 empty=0
time=60 din=33 dout=00 full=0 empty=0
time=80 din=44 dout=00 full=0 empty=0
time=110 din=44 dout=11 full=0 empty=0
time=130 din=44 dout=22 full=0 empty=0
time=150 din=44 dout=33 full=0 empty=0
time=170 din=44 dout=44 full=0 empty=1

```

